Point-driven pencil collector: completed small-size cohorts

Reproducible experiment report

21 October 2026

Scope. Nine exact-order Scholten models, 60 public targets per model: three 16-bit, three 20-bit and three 24-bit source subgroups. Every candidate and same-process source-negation-rho scalar was recomputed on the source curve by the producer. The independent replay and source validator has completed all 540 targets across the nine raw files. The full 21-model, 1,260-target requirement is still open. This document reports measured counts, not a conclusion for the unrun sizes.

Observed full one-target costs

Each pencil solve includes rational-point enumeration, transfer and subgroup checks, point-driven sieve, Jacobian verification, sparse modular elimination, and source verification. The curve setup was physically reused for the 60 targets but its entire measured cost is charged to every cold one-target solve. At $M+0.5S+10I$ and 0.1 source-addition equivalents (GAE) per native logical operation:

p	Verified targets	Verified relations	Row updates	Pencil/rho
739	60/60	21,550	6,288,519	12.931178
1,151	60/60	33,703	20,344,734	28.666756
1,187	60/60	34,726	23,112,414	31.080921
3,371	60/60	98,666	442,470,426	246.117110
3,527	60/60	103,256	499,795,794	267.374941
3,931	60/60	114,989	690,243,338	332.187349
11,731	60/60	343,641	17,235,887,587	4,331.361006
11,887	60/60	348,343	17,655,108,982	3,856.238613
12,143	60/60	355,615	19,113,208,311	4,250.094923

Across the nine models and 36 price cells each, 0/324 pencil cells are below matched rho; cell ratios range from 7.870218 to 5,104.745253. The independent validator recomputed candidate, rho and ratio in all 324 cells from raw rows.

Accounting. The direct prime-field modular arithmetic is recorded separately from the Jacobian backend. For each base-field inversion call, multiplications internal to its Fermat exponentiation are removed from the direct-multiplication count when the I price is applied; scalar modular arithmetic and one sparse-map update per recorded row update are charged at the declared native rate. All 36 pricing cells per model are in `results/pencil-pilot/price-grid.csv`. The same-process rho counts and native counters match the earlier frozen rho controls on every target.

The F739 port reproduces all 61 earlier setup/target records after removing wall-clock fields. The sparse-row collector then recovered all 540 public scalars across these nine models. The raw JSONL and Rust analyzer are under `results/pencil-pilot/` and `src/bin/analyze_pencil.rs`. One algorithm seed per target was used; no uncertainty interval or qualified wall-speed measurement was made.

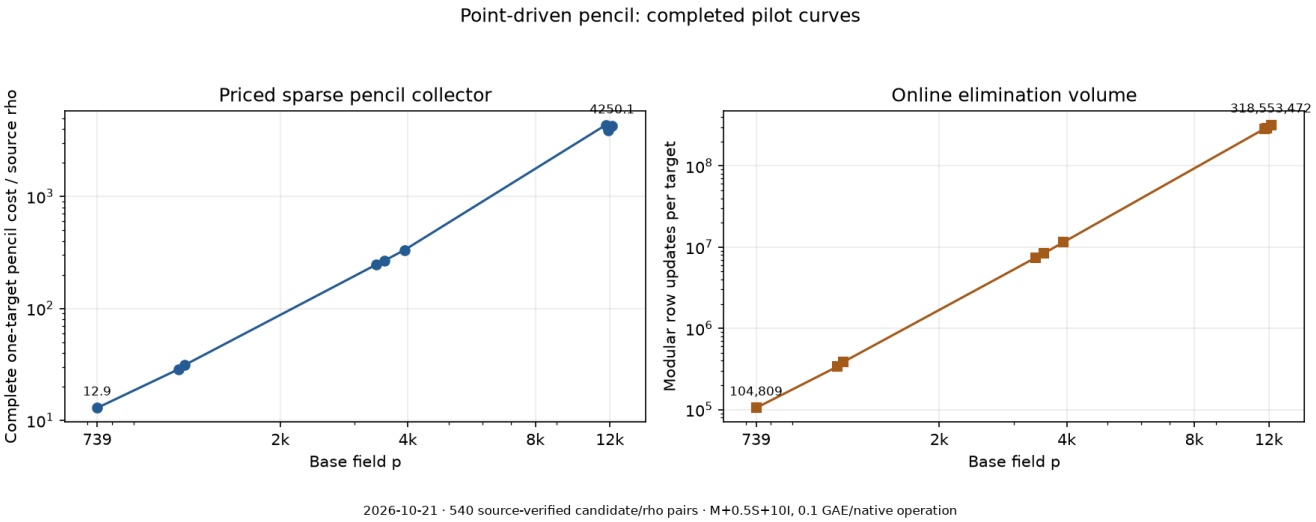


Figure. Complete cold one-target pencil/rho ratio and modular row updates per target on the nine fully replayed curve slices. Values are from the native summary; one seed was used per target and no uncertainty interval was measured.